Artificial Jagged Intelligence: Energy Distribution and the Structural Dynamics of AI Capability Formation

From Gradient Energy Allocation to Optimization Governance

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Abstract

Artificial Jagged Intelligence (AJI) describes the uneven emergence of capabilities in large AI models. This paper argues that AJI arises from uneven gradient energy distribution during optimization. Using the Energetic Paradigm, we interpret capability formation as energy allocation within the training system and propose energy monitoring, redistribution architectures, and optimization governance as a framework for directing AGI development.

A. Artificial Jagged Intelligence and Uneven Capability Emergence

The Emergence of Artificial Jagged Intelligence

In recent years, large-scale artificial intelligence models have demonstrated remarkable progress in several domains, including code generation, mathematical reasoning, language modeling, and strategic search. At the same time, these systems remain fragile in other domains, particularly those involving long-term planning, consistency maintenance, value stability, and real-world situational understanding.

This uneven capability profile has become increasingly evident as models grow in scale. Performance improvements do not occur smoothly across all abilities. Instead, the development of capabilities tends to exhibit irregular jumps: certain dimensions improve dramatically, while others stagnate.

This phenomenon can be described as **Artificial Jagged Intelligence (AJI)**. The term "jagged" refers to an uneven and irregular capability landscape in which localized spikes of performance appear without corresponding global structural upgrades. The model may demonstrate exceptional performance in specific tasks while lacking coherent integration across its broader capability structure.

Importantly, AJI should not be interpreted as a temporary anomaly or measurement artifact. Rather, it reflects a structural property of how contemporary machine learning

systems are trained and optimized. The central question therefore becomes: **why do abilities emerge unevenly in large-scale AI systems?**

Optimization Dynamics and Uneven Capability Formation

A common explanation attributes capability differences to variations in training data. According to this view, models acquire stronger abilities in domains where data is dense and well-structured. While this explanation captures part of the empirical observation, it does not address the underlying structural mechanism responsible for uneven ability formation.

The deeper issue concerns how optimization dynamics allocate learning resources during training. Modern machine learning systems are trained by minimizing an objective function through gradient-based optimization. Parameter updates occur along directions in parameter space that most rapidly reduce the loss function.

Formally, the gradient indicates the direction in parameter space in which a small change produces the greatest reduction in loss. One may interpret this process metaphorically as movement across a landscape: the model occupies a position on a loss surface, where height represents prediction error. The gradient indicates the direction of steepest descent. Optimization therefore proceeds not toward the most balanced or globally coherent direction, but toward the direction that yields the fastest reduction in error.

This optimization dynamic has important implications. If improving a particular capability dimension produces a large decrease in loss, the optimization process will allocate disproportionate update energy to that dimension. Conversely, if a capability has little direct influence on the loss function, its gradient signal will be weak. In such cases, training updates will rarely move in that direction, and the capability will remain underdeveloped.

Consequently, uneven abilities do not primarily arise from bias in the data itself. Rather, they emerge from the structure of the **optimization field**. Capability development reflects how gradient energy is distributed across the model's representational dimensions.

From this perspective, AJI is not best understood as uneven intelligence. It is more accurately described as **uneven optimization energy distribution**.

Gradient Energy and Capability Concentration

To understand this mechanism more precisely, consider the common training objective used in large language models: predicting the next token in a sequence. Under this

objective, gradient updates primarily strengthen representations that directly reduce prediction error.

Certain representational structures are highly effective in reducing token-level prediction loss. These include syntactic patterns, language regularities, programming conventions, and frequently occurring reasoning templates. Because these structures are stable, compressible, and widely present in training data, the gradients associated with them are strong and consistent. As a result, optimization energy naturally concentrates in these dimensions.

In contrast, other abilities—such as long-term planning, value alignment, or cross-context consistency—have little direct impact on token-level prediction accuracy. These capabilities often require multi-step reasoning or extended temporal coherence, which the next-token objective does not explicitly reward. Consequently, the gradient signals associated with these dimensions are weak, and the model allocates very little update energy to them during training.

This asymmetry leads to a characteristic pattern: abilities grow rapidly along directions associated with strong gradients, while other capabilities evolve slowly or remain largely undeveloped. The resulting capability landscape is irregular and discontinuous, producing the jagged structure characteristic of AJI.

B. Energetic Interpretation of Capability Formation

The Energy Landscape of Optimization

Viewed through the lens of the **Energetic Paradigm**, this phenomenon can be interpreted in more fundamental terms. A machine learning model can be understood as an **energy optimization system**, in which gradients represent the direction of energy flow across the parameter space.

In any optimization system, energy tends to move along paths of least resistance. Dimensions characterized by dense data, stable gradients, and clear feedback become low-impedance regions of the optimization landscape. These regions effectively form **energy channels**, directing a large share of gradient updates toward specific representational structures.

Once such channels are established, they reinforce themselves. Strong gradients attract more update energy, further strengthening the associated representations. In contrast, dimensions lacking stable gradients become high-resistance regions of the landscape. Without consistent feedback signals, optimization energy rarely flows into these regions, causing them to stagnate.

From this perspective, AJI is not a defect or anomaly in artificial intelligence systems. It is the **natural consequence of an uneven energy landscape** created by the interaction between data distributions, optimization objectives, and gradient dynamics.

Limitations of Existing Explanations

The uneven development of AI capabilities has been widely observed by researchers and technology companies. These observations are often described using concepts such as *emergent abilities*, *phase transitions*, or *nonlinear scaling effects*. While these descriptions capture aspects of the phenomenon, they typically remain descriptive rather than explanatory.

Several research directions attempt partial explanations. One approach emphasizes **optimization bias**, suggesting that gradient descent tends to learn structures that are easiest to capture first. In this framework, models converge initially toward low-complexity representations or directions with strong gradient signals. Although this perspective acknowledges learning dynamics, it usually treats the phenomenon as a transient property of training rather than a structural determinant of capability distribution.

A second line of research focuses on **scaling laws and phase transitions**. Empirical studies have shown that certain abilities appear abruptly once model size or data scale exceeds specific thresholds. However, these studies typically analyze performance curves and empirical scaling relationships without addressing why particular abilities correspond to steep gradient fields in the first place.

A third research direction examines the **geometry of the loss surface**, analyzing curvature and gradient properties in high-dimensional parameter spaces. While this work contributes to understanding convergence behavior and generalization performance, it rarely connects these geometric properties to the structural distribution of capabilities within the model.

Collectively, these research programs tend to divide the problem into separate components: representation capacity, learning dynamics, and task performance. Few attempts integrate these perspectives into a unified framework explaining how the **global distribution of abilities** emerges from the **distribution of gradient energy**.

In other words, existing studies often describe local mechanisms but do not elevate these insights into a comprehensive theory of capability distribution.

The Energetic Paradigm Perspective

The Energetic Paradigm offers a structural interpretation of this phenomenon. According to this framework, the uneven emergence of capabilities in AI systems reflects the uneven distribution of gradient energy within the optimization field.

Capability spikes occur where energy concentrates; capability gaps emerge where energy allocation remains insufficient. This perspective treats the capability landscape not as a mysterious property of emergent intelligence but as a direct consequence of how optimization resources are distributed during training.

By reframing capability development in terms of energy flow, the Energetic Paradigm provides a global structural explanation for AJI. Rather than describing isolated phenomena, it interprets the jagged pattern of AI abilities as the predictable outcome of an optimization system operating within an uneven energy landscape.

C. Energetic Derivation and Optimization Governance

Analytical Derivation of the Energetic Paradigm

A natural question arises regarding the origin of the Energetic Paradigm: how was this perspective developed, and does it rest on intuitive speculation or on a logically derivable analytical structure? The Energetic Paradigm did not emerge as a single theoretical proposal introduced by a particular researcher or organization. Rather, it developed gradually through sustained analytical engagement with large-scale AI systems and their optimization dynamics. Over time, this interaction produced a convergent structural interpretation of how capabilities emerge during training.

The central methodological principle underlying the Energetic Paradigm is **systematic reduction**. Instead of treating capabilities such as reasoning, planning, or decision-making as primary explanatory units, the analysis repeatedly decomposes these phenomena into progressively lower operational layers.

For example, when analyzing model capabilities, the analysis does not stop at the level of "abilities." When discussing reasoning, the investigation does not remain at the level of reasoning behavior. Likewise, planning is not examined solely as a behavioral outcome. Instead, each phenomenon is traced downward to the processes that produce it. This process follows a sequence of analytical questions:

1. How are these capabilities optimized during training?
2. Through what mechanisms does optimization occur?

3. How are gradients distributed across the parameter space?

4. What resources are consumed and allocated during each update?

By repeatedly applying this downward decomposition, a consistent structural observation emerges. The formation of high-level capabilities ultimately reduces to patterns of **resource flow and allocation** during optimization. In this context, "resources" refer to computational updates, gradient accumulation, and the reinforcement or suppression of representational directions within the model.

This interpretation is not metaphorical. During training, computational resources are physically expended. Gradients accumulate across parameters, and weight updates strengthen some representational directions while suppressing others. Each update step effectively reallocates optimization resources across different dimensions of the model's internal structure.

When the problem of capability formation is reduced to this level, **energy distribution** becomes the most fundamental operational unit of analysis. At this level of abstraction, many higher-level concepts—such as reasoning ability, planning ability, or strategic capability—can be understood as emergent outcomes of how optimization energy is distributed across the model's parameter space.

From this perspective, the Energetic Paradigm is not an external theoretical addition to machine learning. Rather, it is what remains when the analysis reduces the system to its lowest operational layer.

Reduction Principles and the Energy Interpretation of Optimization

The methodological approach described above parallels a recurring pattern in the development of the physical sciences. Many major scientific advances have occurred when complex phenomena were reduced to more fundamental quantities.

For example, classical mechanics interprets force in terms of momentum change. Thermodynamics explains heat as the collective motion of molecules. Chemical reactions can be understood as rearrangements of electronic structure. Information theory ultimately reduces information to binary units. In each case, progress emerged when explanatory frameworks moved beyond surface descriptions and identified deeper operational primitives.

A similar reduction can be applied to artificial intelligence systems. The vocabulary commonly used in AI—terms such as intelligence, capability, reasoning, planning, or value alignment—belongs to a high-level descriptive layer. However, if the analytical process continues downward, the operational activities of the model become clear.

At the training level, the system performs parameter updates. These updates are driven by gradient flow and evaluated through reductions in loss. If the analysis proceeds further, these processes can be interpreted in terms of **resource allocation during optimization**.

Every update step distributes computational resources. Error signals amplify certain representational structures while suppressing others. As a result, some directions within the model's parameter space become progressively strengthened, while others remain weak.

Under this interpretation, capabilities are not "constructed" directly. Instead, they appear as **byproducts of concentrated optimization energy**. When sufficient training energy accumulates in a particular representational direction, the associated capability emerges and rapidly improves.

This observation forms the conceptual starting point of the Energetic Paradigm.

Energy as the Fundamental Operational Unit

Within this framework, the Energetic Paradigm asks a single fundamental question: what is the smallest operational unit that governs capability formation in learning systems?

Traditional analyses often emphasize tokens, parameters, neurons, or architectural modules. However, these elements do not fully capture the dynamic process through which capabilities evolve during training. The Energetic Paradigm instead proposes that the most fundamental operational unit is **energy flow within the optimization process**.

In this interpretation:

- The **gradient** determines the direction of energy flow.
- The **loss surface** defines the energy landscape in which optimization occurs.
- **Computational resources** provide the energy supply driving the system's updates.

Once optimization is interpreted in this way, several empirical phenomena in large-scale AI systems become easier to explain.

First, capability jumps occur when optimization energy suddenly concentrates along a particular representational direction. Second, uneven capability development arises because the gradient field itself is uneven, causing energy to flow preferentially into certain dimensions. Third, scaling-related phase transitions occur because increases in

model capacity alter the topology of the energy landscape, opening new channels through which optimization energy can flow.

These observations follow from a reduction principle rather than from ad hoc theoretical speculation.

Energy Conservation and Capability Allocation

The reduction to energy flow also reveals an important constraint governing capability development. In physical systems, energy is conserved. A comparable constraint exists in computational systems: the available training resources—compute, time, and gradient updates—are finite.

Consequently, allocating optimization energy to one representational direction necessarily reduces the resources available for other directions. Strengthening one capability therefore entails an opportunity cost in other dimensions of the model.

This relationship implies that **energy allocation effectively determines capability allocation**. Once this structural constraint is recognized, the uneven capability profile observed in AJI becomes a natural consequence of how optimization energy is distributed.

Rather than being a mysterious phenomenon, AJI reflects the underlying structure of the energy landscape governing model training.

D. Toward an Energy Redistribution Architecture

Recognizing the energetic structure of capability formation leads to a practical question: can the distribution of optimization energy be intentionally modified?

If the jagged capability profile of AI systems results from the concentration of optimization energy, it becomes theoretically possible to design architectures that redistribute this energy more deliberately. The goal of such redistribution is not to enforce uniform performance across all capabilities, nor to suppress strong abilities. Instead, it aims to achieve three structural objectives.

First, redistribution can prevent excessive amplification of individual capability dimensions. When a single representational direction absorbs a disproportionate share of update energy, it can create structural bias within the system and destabilize the overall capability profile.

Second, redistribution can strengthen dimensions that are systematically neglected during standard training. Capabilities such as long-term planning, cross-context

consistency, and value stability often receive weak gradient signals under conventional objectives. Without explicit intervention, these capabilities remain underdeveloped.

Third, redistribution can help maintain a stable energetic configuration within the optimization field. Excessive concentration of energy in specific regions leads to sharp capability spikes and the jagged patterns characteristic of AJI. While such spikes are not inherently problematic, persistent imbalance may push the system toward extreme representational configurations.

For these reasons, the goal of energy redistribution is not uniformity but **structural stability**. The objective is not to suppress strong capabilities but to avoid systemic imbalance in the optimization landscape.

Optimization Objectives and Energy Regulation

Implementing an energy redistribution architecture requires modifications to the optimization objective itself. Most contemporary AI models minimize prediction error alone. Under this objective, the optimization process has no incentive to maintain balanced energy distribution across capability dimensions.

If a secondary objective were introduced—one that penalizes excessive concentration of optimization energy—the gradient field would change accordingly. Such an objective could encourage the model to reduce both prediction error and structural imbalance in its capability distribution.

Several mechanisms could implement this idea at the architectural level. Examples include dimension-load regularization, energy concentration penalties, representational balance constraints, and inter-module energy quota mechanisms. Each of these approaches would alter how gradient energy flows through the model during training.

In practical terms, such mechanisms would cause the model to incur an optimization cost when too much training energy accumulates in a single capability dimension. The system would therefore be encouraged to distribute updates more broadly across the representational space.

In this sense, the model would no longer optimize solely for performance. It would also regulate the distribution of its internal optimization energy.

This concept represents the foundational idea behind **energy redistribution architecture** within the Energetic Paradigm.

Engineering Implementation of Energy Redistribution

The theoretical analysis presented above suggests that Artificial Jagged Intelligence (AJI) arises from uneven distributions of gradient-driven optimization energy. If this interpretation is correct, the problem becomes an engineering question: can this distribution be directly measured and regulated during training?

A practical implementation begins with monitoring how gradient energy is allocated across the model's internal modules during each training update. Rather than observing only the aggregate loss value, the training process can record the magnitude of gradient updates associated with different functional modules or representational subsystems.

This monitoring process forms a **Gradient Energy Monitor layer**, which measures the relative share of gradient magnitude allocated to each module. Formally, the energy proportion associated with module $i$can be defined as

$$E_i = \frac{\| \nabla_{\theta_i} L \|}{\sum_j \| \nabla_{\theta_j} L \|}$$

Here, Ldenotes the loss function, $\theta_i$represents the parameters belonging to module i, and $\nabla_{\theta_i} L$denotes the gradient of the loss with respect to those parameters. The numerator measures the magnitude of gradient energy absorbed by module i, while the denominator represents the total gradient magnitude across all modules.

The resulting quantity $E_i$represents the proportion of optimization energy directed toward module $i$during a given training step. Persistent dominance of one module's energy share indicates that the optimization field is tilting strongly toward that representational direction. Importantly, this phenomenon becomes directly observable and quantifiable through the monitoring layer.

Dynamic Regulation of Update Energy

Once gradient energy distribution can be measured, it becomes possible to regulate how optimization energy is allocated across modules. One mechanism involves dynamically adjusting the effective learning rate applied to each module based on its observed energy share.

The parameter update rule can be modified as

$$\theta_i \leftarrow \theta_i - \eta\, w_i\, \nabla_{\theta_i} L$$

where η denotes the base learning rate and w_i represents a module-specific weighting factor that regulates the update strength.

A possible definition of the regulation weight is

$$w_i = e^{-\beta(E_i - \mu)}$$

where E_i is the measured energy share of module i, μ represents the desired average energy distribution, and β controls the sensitivity of the regulation mechanism.

Under this formulation, modules that absorb excessive gradient energy automatically receive a reduced effective learning rate. The module can continue to learn, but it cannot dominate the optimization trajectory. This mechanism effectively introduces an **energy regulator** within the optimization process, preventing the emergence of extreme gradient concentration.

Loss-Level Energy Distribution Constraints

Energy regulation can also be implemented directly within the loss function. In addition to the primary task objective, the training objective can incorporate a term that penalizes excessive imbalance in gradient energy distribution.

The combined objective may be expressed as

$$L_{\text{total}} = L_{\text{task}} + \lambda \operatorname{Var}(E_i)$$

Here, L_{task} represents the primary task loss, $\operatorname{Var}(E_i)$ denotes the variance of gradient energy shares across modules, and λ controls the strength of the penalty.

If gradient energy becomes highly concentrated in a small number of modules, the variance term increases the overall loss. The optimizer is therefore encouraged to redistribute gradient energy more evenly across the model's representational structure. This mechanism directly embeds energy balance into the optimization objective itself.

Structural Objectives Beyond Token Prediction

Another source of capability imbalance arises from the narrow objectives used in current large-scale training regimes. Most modern language models optimize primarily for token-level prediction accuracy. Capabilities such as long-term reasoning, cross-context consistency, and multi-step planning receive weak or indirect training signals under this objective.

To address this limitation, additional structural objectives can be introduced into the loss function. For example,

$$L_{\text{total}} = L_{\text{token}} + \gamma_1 L_{\text{consistency}} + \gamma_2 L_{\text{planning}}$$

where $L_{\text{consistency}}$measures cross-context coherence and L_{planning}evaluates multi-step reasoning success. The coefficients $\gamma_1$and $\gamma_2$control the importance assigned to these objectives.

By introducing these terms, capabilities that previously lacked gradient signals begin to receive explicit optimization pressure. Long-term planning and consistency cease to be accidental byproducts of token prediction and instead become formally optimized objectives.

Optimization Architecture Rather Than Parameter Scaling

Taken together, these mechanisms demonstrate that addressing AJI does not primarily require increasing parameter counts or expanding training datasets. Instead, the core challenge lies in redesigning the **optimization architecture** governing gradient energy allocation.

Four engineering components form the foundation of such an architecture:

1. Continuous monitoring of gradient energy distribution.
2. Dynamic regulation of module update rates.
3. Loss functions that penalize excessive energy concentration.
4. Structural objectives that generate gradients for underrepresented capabilities.

These mechanisms can be implemented within existing deep learning frameworks without requiring fundamental changes to model architectures. The Energetic Paradigm therefore operates not merely as a conceptual framework but as a **design principle for optimization systems**.

E. Governance Structure

From Optimization Analysis to Governance

The introduction of energy monitoring and redistribution mechanisms has implications beyond engineering design. Once gradient energy becomes measurable and controllable, the optimization process itself becomes transparent.

Historically, the internal dynamics of large-scale model training have remained largely opaque. Researchers could observe capability emergence only after it occurred, often describing unexpected behaviors retrospectively. The optimization field functioned as a largely uncontrolled natural process.

However, when gradient energy flows can be monitored and regulated, the optimization field becomes an observable and adjustable system. This shift transforms optimization from a purely technical process into a **governable structure**.

Governance in this context does not mean micromanaging every training update. Instead, it involves defining structural constraints that shape the overall behavior of the optimization process. Examples of such constraints include:

- limiting the maximum proportion of update energy allocated to a single capability dimension,
- maintaining minimum energy allocations for critical capability domains,
- reserving dedicated gradient quotas for long-horizon reasoning or planning tasks.

These mechanisms are not philosophical principles; they are formal design rules embedded directly into the training system.

Optimization Governance and the Emergence of Sovereign AI

Once optimization dynamics become governable, capability emergence ceases to be purely accidental. The development trajectory of AI systems can be intentionally shaped through structured energy allocation strategies.

Capabilities may be encouraged to mature at different stages, and training regimes may allocate energy strategically across different functional domains. Under such conditions, AJI no longer represents unpredictable emergence but becomes a partially controllable design outcome.

At this stage, a deeper question arises: **who determines the rules governing the optimization field?**

Possible actors include model developers, data owners, compute providers, corporations, or national governments. The authority to shape optimization rules effectively determines which capabilities receive priority during training.

This issue transforms AI development into a question of **optimization sovereignty**. The central concern of Sovereign AI is therefore not merely where models are deployed

or who controls the data, but who determines the structure of the optimization field that shapes their capabilities.

Governance as the Core Challenge of AGI

The development of Artificial General Intelligence (AGI) is often framed purely as a technical challenge involving scaling, reasoning algorithms, or architectural improvements. While these elements remain essential, the emergence of large-scale AI systems introduces a new dimension to the problem.

As models grow in scale and exhibit cross-domain generalization, unexpected capability jumps become increasingly frequent. The central issue shifts from whether such systems can be built to **how their developmental trajectory is directed**.

If capability emergence is governed by gradient energy concentration, then the trajectory of AGI depends fundamentally on the structure of the optimization field. This structure, in turn, is shaped by design choices including data distribution, objective functions, computational allocation, and training strategies.

These factors are not natural laws; they are human decisions embedded within engineering systems.

Consequently, the key question is no longer whether AGI will emerge, but **which capabilities will emerge first**. Different optimization structures could prioritize radically different domains, ranging from medical reasoning to financial arbitrage, autonomous weapon systems, or large-scale social manipulation.

If the optimization field remains opaque, the direction of capability emergence becomes unpredictable and difficult to control. In contrast, if gradient energy distribution can be monitored and governed, the developmental trajectory of AGI can be constrained within institutional boundaries.

Technology and Governance in the Future of AGI

This perspective suggests a fundamental distinction between two dimensions of AI development. Technological progress determines the **upper bound of capability**, while governance structures determine the **direction in which capabilities evolve**.

If development proceeds solely through technological scaling, AI systems will become increasingly powerful but remain structurally unpredictable. However, if optimization governance mechanisms are incorporated into training systems, capability growth can be directed and regulated.

Under such conditions, AGI may transition from an unpredictable source of emergent risk into a manageable structural capability embedded within institutional frameworks.

The Energetic Paradigm therefore does not merely reinterpret capability emergence. It proposes a pathway through which the optimization processes underlying AI development can become transparent, auditable, and governable—transforming the training system from a black box into an institutional structure.